

Dataset & Preprocessing (B1)

Dataset partition. The full labeled IMDB corpus (25k train + 25k test) is used. The training set is split 90/10 to yield validation data. The 50k unlabeled reviews are excluded.

Dataset split statistics

Split	Samples	Pos	Neg
Train	22,500	11,250	11,250
Val	2,500	1,250	1,250
Test	25,000	12,500	12,500

Preprocessing pipeline (applied in order):

1. **Lowercase** — unifies case; “The” = “the”
2. **Strip HTML** — removes
 and other tags
3. **Remove punctuation** — retain only [a-z\s]
4. **Lemmatize** (NLTK WordNetLemmatizer) — reduces inflected forms to dictionary roots; e.g. *running* → *run*
5. **Whitespace tokenize** — simple split after cleaning

Why lemmatization over stemming? Lemmatization produces valid dictionary forms (e.g. *better* → *good*) that exist in GloVe/Word2Vec vocabularies, whereas stemming produces phonetic roots (*happi*, *runn*) that are absent from pre-trained embeddings, severely degrading coverage. Sequence length statistics: mean = 233 tokens, median = 195, 95th pctl = 590. Setting $T_{max}=256$ covers 61.8% of reviews without any truncation.

Vocabulary & Embeddings (B2)

Vocabulary is built from training data only (prevents leakage). Tokens with frequency ≤ 5 are discarded; the top-30k remaining tokens are retained. Two special tokens are reserved: <PAD> (idx 0) and <UNK> (idx 1). Final vocabulary size: $|V|=28,431$.

Global hyperparameters

Parameter	Value	Note
$ V $	28,431	after min_freq=5 filter
T_{max}	256	truncate/pad to this length

Parameter	Value	Note
d_{emb}	100	Word2Vec & GloVe dim
Batch size	64	fits all models in 16 GB VRAM

One-Hot. Token $w_i \mapsto e_i \in \{0, 1\}^{|V|}$. No semantic information; orthogonal representations. A linear projection layer $R^{|V|} \rightarrow R^{256}$ is prepended to the RNN, adding $\sim 7.3\text{M}$ parameters.

Word2Vec (skip-gram, Gensim): trained on 22.5k IMDB reviews. Config: $d=100$, window=5, min_count=5, 10 epochs. Captures domain-specific sentiment semantics. Vocabulary coverage: 98.1%.

GloVe 6B-100d (pre-trained): Wikipedia + Gigaword (6B tokens). Tokens absent from GloVe are initialised $U(-0.1, 0.1)$. All embeddings are fine-tuned during training. Coverage: 89.1%.

Embedding method comparison

Method	Dim	Corpus	Coverage
One-Hot	28,431	—	100%
Word2Vec	100	IMDB (5M tok.)	98.1%
GloVe 6B	100	Wiki+GW (6B tok.)	89.1%

Vanilla RNN (B3)

Architecture ($L=2$ layers, $h=256$):

The recurrent update at step t is:

$$h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b)$$

Classification: $\hat{y} = \sigma(W_c \text{Drop}(h_T) + b_c)$.

Hyperparameter justification. $L=2$: single layer underfits; $L \geq 3$ compounds vanishing gradients. $h=256$: sufficient capacity for 256-token sequences without overfitting 22.5k samples. Dropout $p=0.5$ between layers and pre-classifier.

Training config: Adam ($\text{lr}=10^{-3}$, $\text{wd}=10^{-5}$), BCELoss, gradient clip norm=1.0, ReduceLROnPlateau, early stopping patience=3.

Challenges.

Vanishing gradients: BPTT gradient at step 1 involves a product of $T - 1$ Jacobians:

$$\frac{\partial L}{\partial h_1} = \frac{\partial L}{\partial h_T} \prod_{t=2}^T \square W_{hh}^\top \text{diag}(1 - h_t^2)$$

Since $\tanh' \in [0, 1]$, this product shrinks exponentially for $T=256$, making early tokens untrainable. Mitigated by gradient clipping (norm=1.0).

Long-range dependency failure: h_T must summarise all 256 tokens. Early sentiment-bearing phrases are progressively overwritten, directly motivating LSTM (B5) and attention (B6).

RNN-GloVe training log (best epoch = 7)

Ep	TrLoss	TrAcc	ValLoss	ValAcc
1	0.582	69.1%	0.563	70.9%
3	0.423	80.9%	0.431	80.2%
5	0.360	84.8%	0.391	83.6%
7	0.329	86.2%	0.379	84.7%

RNN with Different Embeddings (B4)

All three embedding methods are integrated into the same $L=2, h=256$ RNN. For One-Hot a linear projection $R^{|\mathcal{V}|} \rightarrow R^{256}$ is inserted before the RNN; Word2Vec and GloVe are loaded into `nn.Embedding` and fine-tuned.

RNN results across all embedding methods

Model	Params	Test Acc	Conv. Ep
RNN-OneHot	10.31M	78.24%	9
RNN-Word2Vec	3.08M	83.12%	9
RNN-GloVe	3.08M	84.68%	7

Key observations. GloVe edges Word2Vec (+1.56%) despite lower coverage (89% vs 98%) because its training corpus is $\sim 1200 \times$ larger, producing richer semantic geometry. One-Hot is weakest (+3 $\times$ more params from projection, no prior semantic knowledge) and converges slowest.

LSTM Architecture (B5)

The LSTM replaces the vanilla RNN with a four-gate cell that maintains a separate cell state c_t :

$$\begin{aligned}
f_t &= \sigma(W_f[h_{t-1}; x_t] + b_f) \\
i_t &= \sigma(W_i[h_{t-1}; x_t] + b_i) \\
\tilde{c}_t &= \tanh(W_g[h_{t-1}; x_t] + b_g) \\
c_t &= f_t \odot c_{t-1} + i_t \odot \tilde{c}_t
\end{aligned}$$

The **additive** cell update provides a gradient highway. The gradient of c_t w.r.t. c_{t-k} is

$\prod_{j=t-k+1}^t f_j$, which the forget gate can hold near 1 to preserve long-range signals — directly solving the vanishing gradient problem that plagues vanilla RNNs.

LSTM vs RNN: head-to-head comparison (GloVe)

Aspect	Vanilla RNN	LSTM
Gradient flow	Multiplicative (vanishes)	Additive (preserved)
Long-range memory	Poor	Good
Params (GloVe)	3.08M	3.61M (+17%)
Convergence	7 epochs	6 epochs
Test accuracy	84.68%	90.04%
Generalisation gap	≈5.4%	≈3.8%

LSTM results across all embedding methods

Model	Params	Test Acc	vs RNN
LSTM-OneHot	10.92M	82.04%	+3.80%
LSTM-Word2Vec	3.61M	87.92%	+4.80%
LSTM-GloVe	3.61M	90.04%	+5.36%

LSTM-GloVe exhibits a smoother loss curve, converges 1 epoch earlier, and achieves a +5.36% accuracy gain over RNN-GloVe, confirming that gated cell memory is essential for 256-token sequence classification.

Attention-Augmented Models (B6)

Feasibility and Mechanism

Attention is **fully feasible** with both RNN and LSTM. Instead of relying solely on h_T , Bahdanau additive attention aggregates all hidden states:

$$\begin{aligned}
e_t &= v^\top \tanh(W_a h_t + b_a) \\
\alpha_t &= \text{softmax}(e_t), e_t \rightarrow -\infty \text{ if } x_t = \text{PAD} \\
c &= \sum_{t=1}^T \alpha_t h_t \\
\hat{y} &= \sigma(W_c [c; h_T] + b_c)
\end{aligned}$$

The context c is concatenated with the final hidden state h_T before the classifier, combining global selective attention with local recency-weighted information.

Critical Detail: Padding Mask

Without masking, for a 30-token review padded to 256, roughly $226/256 \approx 88\%$ of attention weight falls on <PAD> tokens. Setting $e_t = -\infty$ for padding positions guarantees $\alpha_t = 0$ at all padded positions after softmax.

Computational Overhead

Retaining all T hidden states requires $O(B \cdot T \cdot h)$ extra memory. For $B=64, T=256, h=256$ in FP32: ≈ 1.07 GB additional VRAM.

Attention model results vs baselines (GloVe embeddings)

Model	Params	Test Acc	Time/Ep
RNN-GloVe (base)	3.08M	84.68%	21s
LSTM-GloVe (base)	3.61M	90.04%	23s
Attn-RNN-GloVe	3.41M	88.84%	31s
Attn-LSTM-GloVe	3.94M	92.28%	34s

Attention-LSTM achieves the best accuracy (92.28%) at $1.5\times$ the compute cost of base LSTM. Attention-RNN (88.84%) nearly closes the gap with base LSTM (90.04%), demonstrating that selective focus compensates partially for the lack of gated memory.

Interpretability

Attention weights α_t provide token-level explanations. For the review *“This film is an absolute masterpiece. The acting is superb and deeply moving.”*, top attended tokens are: *masterpiece* (0.182), *superb* (0.157), *moving* (0.123) — all sentiment-bearing words, confirming the model attends meaningfully.

Summary of All Results

Complete results: all models and embeddings

Model	Embedding	Params	Test Acc
RNN	One-Hot	10.31M	78.24%
RNN	Word2Vec	3.08M	83.12%
RNN	GloVe	3.08M	84.68%
LSTM	One-Hot	10.92M	82.04%
LSTM	Word2Vec	3.61M	87.92%
LSTM	GloVe	3.61M	90.04%
Attention-RNN	GloVe	3.41M	88.84%
Attention-LSTM	GloVe	3.94M	92.28%

- **Embedding quality** is the largest single accuracy driver: One-Hot → GloVe adds +6–8 % across all architectures.
- **LSTM** ↗ **RNN** by +3.8–5.4%: additive cell state resolves vanishing gradients and enables long-range dependency capture.
- **Attention** adds +2–4 % over respective base models with $\sim 1.5 \times$ compute overhead; also yields interpretable token importance scores.
- **Best model:** Attention-LSTM + GloVe = **92.28%** test accuracy.
- Accuracy progression: RNN-OH (78.24%) → RNN-GloVe (84.68%) → LSTM-GloVe (90.04%) → Attn-LSTM-GloVe (92.28%).

References. [1] Maas et al., *Learning Word Vectors for Sentiment Analysis*, ACL 2011. [2] Hochreiter & Schmidhuber, *Long Short-Term Memory*, Neural Computation, 1997. [3] Pennington et al., *GloVe: Global Vectors for Word Representation*, EMNLP 2014. [4] Bahdanau et al., *Neural Machine Translation by Jointly Learning to Align and Translate*, ICLR 2015.